

FAI and the Instinct/Reasoning Separation

Derivation Note D2.40 — Note #535 in the CKS Defensive Publication Series

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Full Aspect Integration (FAI), the canonical inter-Self coordination operation in Paper 3 of the CKS theory series, operates entirely within the reasoning-layer domain. All exchange content — DNA-layer specifications and action-layer records — is reasoning-layer substrate content. No instinct-layer content crosses the inter-Self perimeter during a FAI event, and no FAI mechanism touches any participating Self’s instinct layer. This note formalizes that commitment as Derivation Note D2.40, deriving it from D1.10 (exchange bounded to DNA and action layers), D1.19 (instinct non-crossing), and D2.05 (AI-as-mediator at inter-Self scope), with foundational inheritance from Paper 2’s instinct/reasoning separation (B0.01). The note covers the full domain mapping at the inter-Self perimeter, the behavior of LLMs contributing to shared-substrate coordination tasks, the structural mechanism by which the separation preserves each participating Self’s instinct-layer independence, the reversibility property the separation enables, and the instinct cross-contamination anti-pattern that closes the adversarial gap.

1. D2.40 Stated: The Operational Synthesis

D2.40 formalizes the following operational commitment:

At the inter-Self perimeter, the instinct/reasoning separation established by Paper 2 Claim 1 (B0.01) holds without modification. FAI is a reasoning-layer operation throughout. DNA-layer and action-layer content from contributed aspects cross the perimeter within exchange bounds. Instinct-layer content — harness substrates, fast-pattern governing specifications, and anything governing LLM behavior at the instinct scope — does not cross. LLMs contributing to shared-substrate coordination tasks during a FAI event operate in reasoning-layer mode exclusively. Each

participating Self's instinct layer continues to function independently within its home perimeter, unaffected by the FAI event.

This commitment is the inter-Self scope application of B0.01 and the operational elaboration of D1.19. It is not a new architectural claim; it is the articulation of what Paper 2's foundational separation produces when the coordination architecture of Paper 3 is applied at the inter-Self boundary. The prior-art value of D2.40 is the explicit naming and delimitation of the reasoning-layer domain at FAI scope — a clarity that distinguishes FAI from inter-organizational AI coordination mechanisms that involve model fine-tuning, weight transfer, federated averaging of model weights, or any other mechanism by which instinct-layer content influences another organizational boundary's instinct layer.

2. FAI Operates Entirely in the Reasoning-Layer Domain

Paper 2 Claim 1 (B0.01) separates fast-pattern instinct behavior — governed by harness substrates inside the LLM — from substrate-governed reasoning behavior, which lives in the persistent, human-governed CKS substrate outside the LLM. The instinct layer is the LLM operating as a System-1 analogue: pattern-matched, fast-path, no-explicit-reasoning-required responses. The reasoning layer is the CKS substrate operating as a System-2 analogue: explicit, deliberate, human-governed processing encoded in structured substrate content.

When Paper 3 constructs the shared substrate as the medium of coordination between two or more Selves, it extends this separation to the inter-Self boundary. The shared substrate is itself a CKS substrate — carrying all six Paper 1 architectural commitments within its scope — with the distinctive property that its governance perimeter spans more than one Self's home governance perimeter. What it carries is exclusively reasoning-layer substrate content. What it does not carry is anything from the instinct layer.

This means all FAI exchange content falls into one of two reasoning-layer categories:

DNA-layer content. When a Self contributes aspects to the shared substrate, the DNA-layer content of those aspects — orchestration substrates, behavior substrates, schemas, coordination rules, content-domain specifications — crosses the inter-Self perimeter. This content is authored governance content: it specifies how the reasoning layer is governed, what the LLM is authorized to do on the substrate's behalf, and how coordination operates. DNA-layer content is substrate content authored by humans under human authority. It is reasoning-layer substrate content throughout.

Action-layer content. The action-layer content of contributed aspects — recorded task instances, operational outputs, execution records, lived experience accumulated as substrate records — also crosses the inter-Self perimeter within exchange bounds. This content is the reasoning layer's recorded operational history: what was done, under what governance, with what result. It is the output side of the reasoning layer's operation, persisted as substrate content. It is reasoning-layer substrate content throughout.

Nothing from below the reasoning layer crosses. LLM weights do not cross. Harness substrates — the fast-pattern governing specifications that operate inside the LLM — do not cross. Any specification that directly governs the LLM's inference behavior at the instinct scope does not cross. The exchange bound established by D1.10, which restricts exchange to DNA-layer and action-layer content, is the operational form of this commitment.

3. LLMs in the Shared Substrate: Reasoning-Layer Mode Exclusively

During a FAI event, AI operations within the shared substrate conform to D2.05's AI-as-mediator commitment at inter-Self scope: an LLM contributes to coordination tasks by reading from and writing to the shared substrate under orchestration rules, with the substrate as the coordination medium and human authority governing the rules under which the LLM operates.

Any LLM contributing to shared-substrate coordination tasks during a FAI event operates in the following configuration:

Reads are reasoning-layer reads. The shared substrate contains only reasoning-layer substrate content. When an LLM reads from the shared substrate to perform a coordination task, it reads reasoning-layer substrate content. It does not read any participating Self's instinct-layer harness substrates. It does not access the fast-pattern governing specifications of any contributing Self.

Writes are reasoning-layer outputs. When an LLM produces output within a shared-substrate coordination task, that output is substrate content: a contribution to the reasoning layer's coordination record. It is not a model-internal output that bypasses the substrate; it is substrate-mediated output governed by orchestration rules. The LLM's writes during a FAI event are substrate contributions, subject to human authority and inspection, not model-internal state changes.

No instinct-layer access. The LLM does not access any participating Self's instinct-layer harness substrates during the FAI event. The harness substrates that govern LLM behavior within each participating Self's home perimeter are not part of the shared substrate. They are not contributed. They do not cross. They are not readable by an LLM operating in the shared substrate. The LLM's coordination work within the shared substrate is fully bounded by what the shared substrate contains, which is reasoning-layer content only.

This configuration is not a restriction on LLM capability. It is the natural operational posture of an LLM acting as a substrate mediator in a CKS coordination context. The LLM brings its high-dimensional reasoning capability to bear on the substrate content the shared substrate holds; what it does not do is reach across the inter-Self perimeter into any home-perimeter instinct-layer content.

4. The Separation Preserves Home Instinct Independence

The most significant operational consequence of the instinct/reasoning separation at FAI scope is that each participating Self's instinct layer continues to operate independently within its home perimeter during and after the FAI event.

A Self's instinct layer is governed by its harness substrates — the fast-pattern specifications that govern how the Self's LLM behaves across the range of its operational tasks. These harness substrates are not contributed to the shared substrate during a FAI event. They do not cross the inter-Self perimeter. They are not visible to other participating Selves. They are not modified by the FAI event.

The separation produces this independence through three structural properties operating together:

Property 1: Instinct-layer content does not cross. Per D1.19, harness substrates and instinct-layer content do not cross the inter-Self perimeter. A participating Self's instinct layer is architecturally isolated from the shared substrate throughout the FAI event. Whatever occurs in the shared substrate — aspect merges, conflict surfacing, coordination outputs — does not alter the instinct layer of any participating Self.

Property 2: Reasoning-layer outputs from the FAI event do not directly modify the instinct layer. Even after the FAI event dissolves and participating Selves ingest evolution outputs through Paper 2's evolution mechanisms at their home perimeters, the instinct layer is excluded from FAI-feed. DNA-layer content feeds DNA evolution, operating on home DNA through directed selection under home governance. Action-layer content feeds action-feedback evolution, operating within home action layers. Instinct evolution takes no FAI input, by architectural commitment. The layer-routing rule at FAI dissolution is three-case explicit: the instinct evolution case receives nothing from FAI.

Property 3: Any instinct-layer modification requires explicit home governance directed selection. Modifications to a Self's instinct layer — changes to the harness substrates governing LLM behavior — require home governance directed selection per the Paper 2 instinct-layer governance architecture. Such selection requires explicit human authority action within the home perimeter. FAI participation does not trigger, authorize, or initiate that selection. The only path from FAI to a Self's instinct layer runs through explicit home governance action, and that path requires human authority to open.

Together these three properties mean that a Self's instinct layer is the same before, during, and after a FAI event, unless the Self's own home governance independently decides to modify it. The FAI event produces no instinct-layer changes by architectural commitment.

5. The Inter-Self Perimeter Boundary Map

The boundary at the inter-Self perimeter can be stated precisely. For each category of content, the classification is:

Content category	Layer	Crosses?
DNA-layer content from contributed aspects (orchestration substrates, schemas, rules, content-domain specifications)	Reasoning layer	Yes — within exchange bounds per D1.10
Action-layer content from contributed aspects (recorded task instances, execution records, operational outputs)	Reasoning layer	Yes — within exchange bounds per D1.10
Orchestration rules within the shared substrate	Reasoning layer	Authored within the shared substrate under joint human authority — not crossed from home, but governed as shared-substrate content
Harness substrates (fast-pattern governing specifications for LLM behavior)	Instinct layer	No — per D1.19
LLM weights	Instinct layer and below	No — per D1.19 and D1.10
Model fine-tuning signals or adaptation gradients	Instinct layer	No — per D1.19

The boundary is clean. Everything above the instinct layer, and nothing at the instinct layer or below, crosses during FAI. The shared substrate contains reasoning-layer substrate content and nothing else. This is the operational form of the instinct/reasoning separation at inter-Self scope.

6. Reversibility: The Separation as the Structural Mechanism

FAI participation is genuinely reversible. When a Self withdraws from a FAI event (per D2.27), the Self returns to its prior governance state. The reasoning-layer substrate at the home perimeter may have received FAI evolution outputs, which continue to operate as input to the home substrate's evolution mechanisms under home governance. But the instinct layer is unchanged from its pre-FAI state.

This reversibility is not merely a governance option — it is architecturally guaranteed by the instinct/reasoning separation. The guarantee works because the instinct layer was never touched. There is nothing to undo in the instinct layer, because nothing in the instinct layer changed. The Self's fast-pattern behavior — the pattern-matched responses its instinct layer

produces — is the same after withdrawal as before FAI participation.

Compare this with an inter-organizational AI coordination mechanism that involves model fine-tuning, weight sharing, or any other form of instinct-layer exchange. In such mechanisms, participation leaves a permanent mark: the model's weights have changed, and reversal requires rolling back the weights or retraining from a prior checkpoint. Reversal is costly, potentially incomplete, and in practice rarely attempted. The instinct/reasoning separation eliminates this cost entirely by structural design: because FAI never touches the instinct layer, withdrawal is complete and does not require any weight-level undo operation.

This reversibility property has governance implications. A Self's governing authority can evaluate FAI participation, engage in the shared substrate for the duration of the event, and withdraw if the event's evolution outputs are not authorized for absorption — without any permanent commitment to the other participating Self's influence on the home instinct layer. The separation makes the governance boundary enforceable at the instinct layer as an architectural fact, not merely as a policy promise.

7. Anti-Pattern: Instinct Cross-Contamination

The instinct cross-contamination anti-pattern is any FAI mechanism — direct or indirect — through which participation in a FAI event leads to changes in a participating Self's instinct layer that were not authorized by explicit home governance directed selection.

The direct form is obvious: any mechanism that allows harness substrates or LLM weights from one Self to influence the instinct layer of another Self during a FAI event is a direct violation of D1.19 and D1.10. No compliant implementation would include such a mechanism.

The indirect form is more subtle and more important to name. The indirect form occurs when:

1. A Self participates in a FAI event and ingests reasoning-layer evolution outputs into its home substrate at dissolution.
2. The absorbed DNA-layer or action-layer content triggers, directly or through automated pipeline logic, a modification to the home instinct layer — for example, by initiating a fine-tuning run or updating a harness substrate specification.
3. This instinct-layer modification occurs without explicit home governance directed selection authorizing it.

This indirect chain violates the commitment even though no instinct-layer content crossed the perimeter. The FAI event influenced the instinct layer through an intermediate reasoning-layer step, bypassing the explicit governance authorization that home governance directed selection requires.

A compliant architecture forecloses this chain structurally. The three structural properties in §4 — instinct content does not cross, reasoning-layer FAI outputs do not directly modify the instinct layer, and any instinct-layer modification requires explicit home governance directed

selection — together ensure that the indirect chain cannot complete without an explicit human authority action at the home perimeter. Automated pipelines that translate substrate content changes into model fine-tuning signals without explicit governance authorization are not compliant with the instinct/reasoning separation at FAI scope.

The anti-pattern closes an adversarial gap that is easy to overlook in practical implementations. Engineering teams building FAI-compatible systems should verify that no automated pathway translates FAI-derived substrate content into instinct-layer modifications, and should treat any such pathway as an architectural violation of the commitment D2.40 formalizes.

8. Inheritance

D2.40 inherits from four sources without redefense:

Paper 2 Claim 1 / B0.01 (instinct/reasoning separation). The foundational separation of fast-pattern instinct behavior from substrate-governed reasoning behavior into independently-evolving layers under unified human governance is Paper 2's first and foundational claim. D2.40 applies it at inter-Self scope: the separation holds at the inter-Self boundary without modification.

D1.10 (exchange bounded to DNA and action layers). The exchange-bounding commitment established in D1.10 restricts what crosses the inter-Self perimeter to DNA-layer and action-layer content. D2.40 inherits this restriction and develops its implication: because only reasoning-layer substrate content is within exchange bounds, FAI is a reasoning-layer operation by construction.

D1.19 (instinct non-crossing commitment). D1.19 established that harness substrates and instinct-layer content do not cross the inter-Self perimeter. D2.40 is the operational elaboration of D1.19 — it covers the full set of consequences the non-crossing commitment produces for FAI operations, LLM behavior in the shared substrate, home instinct independence, and reversibility.

D2.05 (AI-as-mediator at inter-Self scope). D2.05 established that AI operates as a substrate mediator at the inter-Self scope, with the shared substrate as the coordination medium. D2.40 inherits this mediator posture and confirms that LLMs in the shared substrate operate entirely over reasoning-layer substrate content.

The Paper 3 named foil — opaque inter-agent merge of outputs without governable substrate — connects here. An opaque agent-to-agent communication mechanism that crosses instinct-layer content would violate both D1.10 and D1.19 simultaneously: it would exceed the exchange bounds (D1.10 violation) and cross instinct-layer content (D1.19 violation). D2.40's commitment is what makes FAI distinct from that foil.

9. Operational Test

For any implementation claiming FAI compliance with respect to the instinct/reasoning separation, an observer should be able to verify the following:

Test 1. Every element of the shared substrate during a FAI event — every contributed aspect's DNA-layer and action-layer content, every orchestration rule governing the shared substrate, every LLM output produced in the coordination of the event — can be classified as reasoning-layer substrate content. No element is a harness substrate specification, an LLM weight, a model fine-tuning signal, or any other form of instinct-layer content.

Test 2. An auditor examining the shared substrate at any point during the FAI event finds no content that governs the fast-pattern behavior of any contributing Self's LLM. Harness substrate content is home-perimeter content; it is not present in the shared substrate.

Test 3. Every LLM read operation accesses only what the shared substrate contains. Every LLM write operation produces substrate content that is inspectable under shared-substrate governance. No LLM operation reaches into any participating Self's home instinct layer.

Test 4. At the home perimeter, there is no automated pipeline that translates FAI-derived evolution outputs into instinct-layer changes without an explicit human governance directed selection event authorizing the change. The pathway from FAI content to the instinct layer is gated by explicit human authority action, not by automated substrate state monitoring.

A FAI implementation that satisfies all four tests instantiates the instinct/reasoning separation at inter-Self scope as D2.40 formalizes it.

10. Conclusion

D2.40 formalizes how Paper 2's instinct/reasoning separation operates at the inter-Self boundary during FAI events. FAI is a reasoning-layer operation throughout: DNA-layer and action-layer content from contributed aspects are both reasoning-layer substrate content; the shared substrate contains no instinct-layer content; LLMs contributing to shared-substrate coordination tasks operate over reasoning-layer substrate content exclusively; and no FAI mechanism touches any participating Self's instinct layer, directly or indirectly.

The separation produces three architectural properties at FAI scope. First, it isolates each participating Self's instinct layer: the Self's fast-pattern behavior is governed by its home harness substrates, which do not cross the inter-Self perimeter and are not visible to or accessible by the shared substrate. Second, it makes FAI participation genuinely reversible: withdrawal returns the Self to its prior governance state without any weight-level undo operation, because the instinct layer was never modified. Third, it forecloses the instinct cross-contamination anti-pattern: any pathway from FAI participation to instinct-layer modification requires explicit home governance directed selection, architectural gating that automated pipelines cannot bypass.

These three properties together define what it means for an inter-Self coordination mechanism to respect the instinct/reasoning separation — and establish the prior-art commitment against which any future mechanism operating over the instinct layer at inter-Self scope must be evaluated.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

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